A BM TECHNICAL MANUAL

Graphics techniques catalogue

Every 2D and 3D technique of bm on the CPU, the V3D (Pi Zero W) and the Mali-G52 (RGB30): what it is, what is built, what it costs, and what is not built.

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README

What this is. A list, one technique at a time, of what bm does to draw: 3D rasterising, lighting, textures, 2D sprites and maps, light tables, and the work that the GPUs take. Each row says if the technique is **Built**, **Partial**, **Not built** or **Dropped**, and why when it is not built.

How to read it. Three machines: the **ARM** (Pi Zero W, one core; RGB30, four cores), the **V3D** (VideoCore IV of the Pi Zero W) and the **Mali** (Mali-G52 of the RGB30). The last column gives the equivalent in the public APIs (OpenGL ES, Vulkan, Direct3D, OpenVG, OpenCL, OpenVX, SDL2 and SDL3, Mesa) so you can compare.

Where the numbers come from. Benchmark reports of the 3D Bench, 640x360, RGB565 on the Pi and XRGB on the RGB30. Sources: `docs/PERF-BUDGET.md`, `docs/progress/graphics.md`, `docs/progress/graphics-mali.md`, `docs/progress/graphics-vector.md`, `docs/progress/multicore.md`. A number marked *not measured* was never measured. Nothing is invented.

How to read the tables

Word	Meaning
Built	Works in the kernel and has a test.
Partial	Works for some cases. The row says which.
Not built	Nothing in the code. The row gives the reason or says that nobody decided.
Dropped	Tried or planned, then left out. The row gives the reason.

Load figures are "at 60 fps": how many of the thing fit in one frame of 16.6 ms. The Pi Zero W is the reference machine. A RGB30 number has its own note.

3D on the CPU (ARM rasteriser)

The ARM transforms, lights, clips and draws the triangles itself. It is the only 3D path on the RGB30 before the Mali, and the fall-back of both GPUs. Source: `src/bm/r3d.c` .

Technique	State	What it does	Measured	Equivalent
Triangle rasteriser	Built	Edge functions in fixed point. Flat or Gouraud (dithered) fill, one pixel at a time.	Pi: 70.8 spheres of 96 faces at 60 fps. RGB30: 177.3 (old report, bm3d 6.8)	GL fixed pipeline raster; Vulkan `vkCmdDraw`
Depth buffer	Built	16-bit z-buffer in RAM, cleared with DMA.	Part of the figures above	GL depth test; Vulkan depth attachment
Perspective-correct textures	Built	Texels from the sheet with correct divide; loops specialised by case.	Pi: hero with texture 3.4 at 60 fps	GL `texture2D` ; D3D sampler
Near-plane clipping	Built	The ARM cuts triangles on the near plane.	Not measured alone	GL clip volume
Back-face culling	Built	The side where vertices go clockwise is visible.	Not measured alone	`glCullFace`
Light: sun, sky and ground, specular, rim, point lamps	Built	Computed per vertex or per face on the ARM with the VFP.	Part of the figures above	GL fixed lights; D3D lights
Fog	Built	Mix with a fog colour by depth.	Not measured alone	`glFog` (GL 1.x)
Per-face materials	Built	Emissive, glossy, screen-door (checker transparency), LOD.	Not measured alone	Material state in D3D
Planar shadows	Built	A flattened copy of the mesh on a floor plane.	Pi: hero with shadow, ARM misses 60 fps (24.7 ms for 1 hero)	Projection shadows (old GL technique)
Rigid skinning	Built	One bone per vertex; two-bone groups on the GPU.	Pi: hero of 1536 triangles with 16 bones, 3.9 at 60 fps	GPU skinning in a vertex shader
Fast mode (`r3d_fast=1`)	Built	One object-to-camera matrix, light in model space.	2.2 to 3.6 percent fewer ARM instructions per triangle	-

Technique	State	What it does	Measured	Equivalent
Flat textures (flag 128)	Built	A textured face is drawn in its average texel.	Not measured alone	-
Cel shading	Built	Light cut into 4 bands. Option `dither3d` .	Cost against sun light: x1.00 on the ARM (Pi)	Toon ramp in a fragment shader
Outline (inverted hull)	Built	Draws the back faces larger, in a dark colour (flag 512).	x1.34 to x1.50 of the lit hero on the ARM (Pi)	Hull outline (common technique)
Point, line and billboard	Built	Sprites with depth in the 3D scene.	Not measured alone	Point sprites
Impostors (`r3d_impostor`)	Partial	Far heroes drawn as a picture. Off by default: a Pi session at 8 gave no gain.	Pi: 57.7 fps, GPU 7.8 ms a frame at setting 8	Billboard impostors
3D particles	Built	ARM step and draw (`particles*`).	Not measured on the Pi	Particle systems; Compute shader
Sprite stacking	Built	A model made of slices (`stack` , `zstack` , `stackv`). ARM only.	Pi: 54.6 stacks of 8 slices at 60 fps	- (a bm technique)
LOD by distance	Built	Mesh level by size on screen.	Not measured alone	Tessellation / LOD
Shadow maps	Not built	Nobody decided. Planar shadows exist.	-	Shadow mapping (GL, Vulkan, D3D)

The ARM on four cores (RGB30)

Technique	State	What it does	Measured	Equivalent
Faces split among the cores	Built	`offload_par` gives one block of faces to each core, 96 faces or more, flat or baked meshes without texture. A hung core is marked and the caller runs its block.	RGB30 3D Bench ARM score 466 -> 601	Parallel vertex processing

Technique	State	What it does	Measured	Equivalent
Pixels in bands of rows	Built	Each core draws its own rows of the page (`r3d_bands` , flat, Gouraud, textured). Same pixels as one core.	`heroes` 8.0 -> 15.8, `heroes_tex` 6.3 -> 12.7, `clip` 325 -> 692	Tile-based parallel raster
Draw build for the Mali on the cores	Built	`build_draws` in `mali.c` splits the draws among the cores in blocks (32 draws or more). Key `mali_par` .	Games 451 -> 491, GPU 433 -> 513	Multi-threaded command recording (Vulkan)
Cores on the Pi Zero W	Not built	The Pi Zero W has one core. A first try with 4 bands of tables lost the firmware mailbox.	-	-
Lighting per face on worker cores	Not built	Idea Q2 of the Mali queue. The loop is a pure function of the face range.	-	-

3D on the V3D (Pi Zero W)

The V3D is the VideoCore IV (a tile renderer). bm drives it without Linux and without the firmware 3D library. Source: `src/gpu/gpu3d.c`, `v3d.c`, shaders written in `tools/qpuasm.py`. Version history: `docs/DRIVERS.md`.

Technique	State	What it does	Measured	Equivalent
Tile binning and render	Built	The binner makes the tile lists; the renderer draws tile by tile. Two hardware jobs per frame.	Raw: 2966 thousand small triangles a second; 811 Mpixel a second (full screens, render only)	Tile-based GPU (Mesa vc4)
Fragment shaders on the QPU	Built	Flat, Gouraud, textured, screen-door, cel.	`quad_flat`: 163.7 quarter screens at 60 fps (ARM 11.5)	GL fragment shader
Vertex shaders on the QPU	Built	Transform, light, bones, shadows; the GPU places the corners (`gpu3d_vs=2`).	`heroes`: 66.3 at 60 fps (GPU alone 6.3, ARM 3.9); `spheres` 782.9 (ARM 70.8)	GL vertex shader
Coordinate shaders and near clip	Built	A separate shader gives clip coordinates for the binner. The GPU clips on near plane and guard band.	Part of the vertex figures	Vulkan binning pass
Indexed meshes (16-bit)	Built	Shared vertices are set up once. Meshes up to 65535 vertices and faces.	`big`: 9.2 models of 10080 triangles	`glDrawElements`; `vkCmdDrawIndexed`
Depth and early z	Built	24-bit depth in the tile buffer. Early z is off after a `zclear()` and with MSAA.	Not measured alone	Depth test; early-Z
MSAA 4x	Built	Option `gpu3d_aa`. Antialiasing is the only antialiasing in bm.	Not measured in the sheets used here	`GL_SAMPLES`; Vulkan MSAA
Job queue (1 job)	Built	The GPU job starts at the end of the frame. The next `_update` runs meanwhile. Default on the Pi.	`queue` test: 557.8 -> 631.1 spheres (+13 percent)	Async submit (`vkQueueSubmit`)

Technique	State	What it does	Measured	Equivalent
Job queue (2 jobs)	Built	Two job memory blocks; the ARM fills one while the GPU draws the other.	`herotris`: 49956 triangles at 13.7 ms (Q) and 50024 at 13.5 ms (Q2)	Frames in flight
Frame held one frame (`gpu3d_later`)	Built, off	Costs a frame of latency.	+0.4 percent `herotris`, +10 percent `heroes`	-
Draw merging (`gpu3d_merge`)	Built	Many small pieces become one mesh in world axes. Only from 32 pieces.	100 pieces: +24 to +33 percent; 10 big pieces: -26 percent	Instancing / draw batching
Textures RGBA8888 (T-format)	Built	Up to 8 textures in a job.	`quad_tex`: 131.1 quarter screens	`glTexImage2D`
Textures RGB565 and RGBA5551	Built, off	Half the memory and half the texture reads (`gpu3d_tex16`). No gain on the Pi (21 Oct).	-	Compressed or packed formats
Textures ETC1	Built, off	`gpu3d_etc1`, encoder in `src/gpu/etc1.c`. No gain on the Pi.	-	ETC1 (GL ES 2 extension)
Bilinear filter	Built	`gpu3d_filter=1`. The look differs from the ARM. 2D stays nearest.	Fill: 2.3 ns a pixel against 1.65 nearest	`GL_LINEAR`
Compact vertex attributes	Built, off	`gpu3d_attr=1`. `herotris` 54056 plain, 54013 compact: no gain, the frame is ARM-bound.	72.2 heroes against 66.2 in `heroes` only	Packed vertex formats
Hardware sort (`gpu3d_sort`)	Dropped	Off: the numbers show no gain.	`heroes`: 14.2 ms against 15.9 ms but the pixels field was 0	Sorting transparent geometry
Cel shading on the QPU	Built	Light cut into 4 bands in the vertex programs.	x0.99 to x1.02 of sun light; 50 heroes at 12.1 ms	Toon shader
Outline on the GPU	Built	Inverted hull in the vertex programs.	x1.05 to x1.23 of the lit hero; 42.4 heroes at 14.4 ms	-

Technique	State	What it does	Measured	Equivalent
Draw calls	Built	One cube per call.	1386 calls at 60 fps (12 us a call; ARM 467)	Draw-call count
Many triangles at 60 fps	Built	`herotris` with the vertex shader and the queue.	About 50000 triangles at 13.7 ms	-
Compute on the QPU	Built	Programs outside 3D: fade by levels and a particle step (`gpu3d_qpu`). Only `dark_end` uses them.	Fade 8.6 to 9.3 ms against 20.4 ms; particles 1024 in 47 to 52 us against 158 to 171 us	OpenCL kernels; compute shaders
Chained jobs (CT0/CT1)	Not built	Only if a measurement shows the GPU idle between jobs. It is not: the frame is ARM-bound.	-	-
Geometry and tessellation shaders	Not built	The VideoCore IV has no such stages.	-	GL 4 geometry and tessellation
Ray tracing	Not built	No hardware; never considered.	-	`VK_KHR_ray_tracing`

3D on the Mali-G52 (RGB30)

The Mali is a tile renderer of the Bifrost family. bm builds its job descriptors from Mesa (`v7.xml` , MIT) and its shaders with Mesa's Bifrost compiler (`tools/mali/mkshader.c`). Source: `src/rgb30/mali.c` , `mali_desc.c` . Table of parity with the V3D: `docs/progress/mali-parity.md` .

Technique	State	What it does	Measured	Equivalent
Triangle through Vertex, Tiler and Fragment jobs	Built	Checked on the device with a pixel check.	-	Panfrost job chain
Vertex shader (`BM_VS` , `BM_VS3`)	Built	Position in pixels, or a 4x4 matrix pushed per draw (model-space vertices).	`draws` : 533.8 at 60 fps with the shader against 116.5 without	Vertex shader
Fragment shader, flat colour	Built	One colour a draw, pushed to the shader.	`quad_flat` : 143.2 quarter screens (ARM 15.8)	Fragment shader
Per-face light, cel and fog	Built	Computed on the ARM in `be_mesh` , then sent as draws.	The cost is on the ARM: 6960 instructions a triangle in `spheres` , against 1213 in `spheres_unlit`	Per-vertex light
Textures (opaque, nearest)	Built	Texture, Surface and Sampler descriptors, a 2 MB pool, key `mali_tex` . Alpha, screen-door and bilinear stay on the ARM.	`quad_tex` 6.4 -> 126.3; games score 482 -> 530	`glTexImage2D`
Draw batching in one pass	Built	4 MB arena, 1024 draws in one GPU pass; more than that merges in screen space.	Games score 192 -> 247; GPU score 14 -> 228 (bm3d 7.15)	Command buffer
Async job queue (2 sets)	Built	The ARM works while the GPU draws (`gpu3d_queue=1` or `2`).	2 frames in flight finish in 204 us	Frames in flight
Resolve into the page	Built	Cached 2 MB resolve; key colour `0xFF00FF` marks "not covered".	`resolve_us` 0.8 to 2.1 ms in a 14 ms frame	Present / blit
Draw build on the cores	Built	See the four-core table above.	Games 451 -> 491	-

Technique	State	What it does	Measured	Equivalent
Hardware Z-buffer and early z	Partial	In work (track G2 of `docs/progress/multicore.md`). Today the ARM sorts triangles far to near (`sort_tris`).	-	Depth attachment
Lighting per face in the vertex shader	Not built	Queue item Q2. The Pi does it in its vertex shader; the Mali only transforms.	The Pi `spheres` 782.9 against 22.9	Per-vertex lighting
Alpha blend and bilinear on the GPU	Not built	They stay on the ARM.	-	`glBlendFunc`, `GL_LINEAR`
MSAA	Not built	No Mali setting is used.	-	MSAA
2D on the Mali	Dropped for sprites	Each draw costs about 4 us, so 600 sprites cost 2.4 ms, more than the ARM path (about 0.5 ms). Worth it only for big layers and fades, after the texture work.	-	-
Mali beats the ARM raster	Partial	In some scenes the ARM on four cores is ahead: `spheres_unlit` ARM 164.4 against GPU+VS 103.8; `heroes`, `baked`, `shine` and `split` the same. The Mali wins `quad_flat` and `draws`.	See rows above	-

2D on the CPU

The 2D code (`src/bm/gfx16.c`) is the same on the Pi and the RGB30. Pixels are RGB565.

Technique	State	What it does	Measured	Equivalent
Primitives	Built	`line` , `rect` , `rectfill` , `circ` , `circfill` , `tri` (flat or Gouraud with three colours), `pset` , `pget` .	Not measured alone	`SDL_RenderLine` , `SDL_RenderFillRect` , OpenVG `vgDrawPath`
Clip rectangle	Built	`clip` limits every call.	-	`SDL_SetRenderClipRect` , scissor
Sprites	Built	`spr` , `sspr` with zoom and flip; 1-bit alpha (a pixel is on or off).	2574 sprites 16x16 at 60 fps (6.5 us each)	`SDL_RenderTexture`
Tile map	Built	`map()` up to 8 layers, tile flag mask, packed copy of the cells (128 bytes a cell).	One opaque full-screen layer 3.08 ms; 7.7 layers fit 60 fps. Packed pool: 2.9 -> 2.0 ms on Yharnam	Tilemap in a game engine
Bitmap text	Built	Fonts 6x12, 8x14, 8x16, 16x32, 24x48, integer scale.	Not measured alone	`SDL_ttf` bitmap use
Sprite stacking, rotation of a slice	Built	`stack()` rotates and scales a slice (nearest sampling).	54.6 stacks of 8 slices at 60 fps	Affine transform of an image (OpenVG)
Light tables (`fades` , `g16_fade_apply` , `g16_fade_glow`)	Built	Darkness and lamps by a table of colour levels.	Full-screen fade 20.3 ms; light 30.7 ms (Pi); at 30 fps 24.7 and 20.3 lamps	Colour lookup table
Screen modes	Built	`screen(w, h)` from 320x180 to 1920x1080; the HDMI scaler enlarges at no CPU cost.	-	`SDL_SetRenderLogicalPresentation`
Bitmap blit with scale	Partial	Works. It divides for every pixel; the ARM1176 has no divide instruction. A table of x indices would be 2 to 3 times faster (estimate).	Not measured	`SDL_RenderTexture` with scale

Technique	State	What it does	Measured	Equivalent
Text run drawing	Partial	`g16_text` tests every pixel. Drawing runs would be about 2 times faster (estimate).	Not measured	-
Four-core bands for blit, light and `cls`	Not built	RGB30 only, about 3 times on full-screen passes (estimate). The `r3d` band pattern exists.	-	-
Per-file `-O3` for 2D and 3D (`PI_O3_FILES`)	Partial	Built and tested on the host; not measured on the Pi. Accept only with 3 percent less `ms` in `gpu2d` and `mix` and 5 percent more kernel.	-	-
2D benchmarks	Built	`b3d tests=map8,sprites,stack,fade,light,gpu2d` .	See `docs/PERF-BUDGET.md`	-
Sprite rotation with a transform stack	Not built	API gap against LÖVE (API-7). A design note comes first.	-	`SDL_RenderGeometry` ; Vulkan transform
Off-screen render targets	Not built	API gap (API-8 to API-11).	-	`SDL_SetRenderTarget` ; OpenVG `VGImage`
`palt` with alpha, blend modes	Not built	API gap. The only blend is `g16_fade_glow` .	-	`SDL_SetTextureBlendMode` ; OpenVG blend modes

2D on the GPUs

Technique	State	What it does	Measured	Equivalent
2D over the 3D in the V3D job	Built	Rectangles, points, lines, `spr`, `sspr` (whole zooms), `map` cells and `print` text become quads in the same job.	3569 sprites or more at 60 fps (ARM 2574)	Textured quads
Light by the GPU (`gpu3d_light`)	Built	The V3D lights the page that `dark_begin` .. `dark_end` makes. Up to 12 lamps in a tile of 64x64 pixels. The lamps add up; the ARM light has levels.	Yharnam: light 12.9 ms -> 1.7 ms; 31.6 -> 50.1 fps autoplay. Pi only	Deferred light pass
Fade by the QPU (`gpu3d_qpu`)	Built, off	Full-screen fade on the QPU.	8.6 to 9.3 ms against 20.3 ms	Post-process pass
Menu on the V3D (`menu_scale`)	Built	The menu is drawn at 720p or 1080p by the V3D, enlarge cost 0. 1080p verified; 720p not yet.	-	-
Map as a resident batch of quads	Not built	The idea: a block of 16x16 cells is one buffer made once. The GPU 2D today builds each quad again: slower than the ARM on a big map (44.3 fps against 50.5 on Yharnam).	Estimate: 2 to 2.5 ms of gain, not measured	Vertex buffer object
2D only on the GPU (`gpu2d_only`)	Dropped for big maps	Slower than the ARM.	See above	-
Light by the Mali	Not built	No Mali equivalent of the V3D light shader. Considered only if Yharnam runs under 60 fps on the RGB30.	-	-

Technique	State	What it does	Measured	Equivalent
Fade of `dark_end` as a fragment shader	Not built	The colour tables have 65536 entries. Only a scale without a table fits, and the pictures would differ.	-	-
Copy to the page by the GPU	Dropped	DMA takes 3.1 ms against 1.8 ms on the ARM.	-	-

The OpenVG and VGU list (2D vector)

bm does not adopt the OpenVG API: its Pi firmware version needs the VideoCore firmware services, and the Mali has no OpenVG. Each item below would be a function of bm, in the style of `docs/api/draw.md`. Costs are estimates from the code. Source: `docs/progress/graphics-vector.md`. Antialiasing is out of scope.

Technique	State	What it does	Estimate	Equivalent
Solid colour paint	Built	Every draw call.	-	`vgSetColor`
Linear gradient	Partial	`tri` with three colours; `rectgrad` would be two Gouraud triangles.	0.05 ms	`vgSetParameter` gradient
Radial gradient and spread	Not built	Forward differences on the squared distance and a 256-entry table. Priority 1.	About 6 cycles a pixel	OpenVG radial gradient
Path (move, line, quad, cubic, arc, close)	Not built	Subdivide curves, an edge list and scanline fill. Priority 2.	3 ms full screen; 0.02 ms for a 32x32 icon on the Pi	`vgAppendPathData`
Fill rules (even-odd, non-zero)	Not built	Part of the path code.	About 0 extra	`VG_EVEN_ODD`
Stroke (width, caps, joins, miter, dash)	Not built	Offset quads and round joins. Only 1 pixel lines exist.	-	`vgDrawPath(VG_STROKE_PATH)`
VGU shapes (round rect, ellipse, arc, polygon)	Not built	`rrect`, `ellipse`, `poly` on top of `circfill` and `rectfill`. Priority 1.	`rrect` 0.03 ms	`vguRoundRect`, `vguEllipse`
Affine transform of an image	Partial	`stack` rotates. `sspr_affine` with a 2x2 matrix would use `stack_core`. Priority 1.	16x16: 0.01 ms (Pi)	`vgDrawImage` with a matrix
Perspective warp (VGU)	Not built	Two textured triangles. Priority 3.	0.02 ms for 16x16	`vguComputeWarpQuadToQuad`
Image formats with real alpha	Partial	Today alpha is on or off. A coverage byte 2-255 in the sheet would work with `blit_alpha`.	About 8 cycles a pixel more where alpha is not binary	ARGB images

Technique	State	What it does	Estimate	Equivalent
Blend modes	Not built	`blend(mode)` : alpha, add, multiply, screen.	Full screen 2 ms (Pi); 0.5 ms with bands (RGB30)	`VG_BLEND_*` ; `SDL_BlendMode`
Mask layers	Partial	Tile flags only.	One more load a pixel	`vgMask`
Pattern paint	Partial	`spr` and `map` cover most uses.	-	Pattern paint
Colour matrix, convolve, Gaussian blur	Not built	Box blur with running sums. Blur 320x180: 1.2 ms (Pi); 0.1 ms with bands (RGB30).	See text	`vgColorMatrix` , `vgGaussianBlur` ; OpenVX filters
Vector fonts	Not built	`stb_truetype` (public domain), glyphs cached as an alpha sheet.	0.05 to 0.1 ms a glyph; cached blit 0.01 ms	`vgDrawGlyph` ; `SDL_ttf`
Antialiasing of 2D shapes	Dropped	Out of scope (user, 28 Oct).	-	`VG_RENDERING_QUALITY`

Order of work for 2D, agreed with the user: the "better" items of the existing code first, then `rectgrad` , `rrect` , `ellipse` ; then `sspr_affine` and alpha; then paths and stroke; then vector fonts and blur. The Mali takes 2D only for big layers.

What the other APIs have that bm does not

API	Feature	bm
OpenGL ES 2 / Mesa vc4	Programmable shaders on the VideoCore IV	Built, with own QPU assembler (`tools/qpuasm.py`), no OpenGL
Vulkan	Command buffers, explicit sync, several queues	Partial: a job queue with two sets, no general command API
Direct3D	Feature levels, shader models	Not applicable
OpenCL	General compute	Partial: QPU programs for the fade and particles. No OpenCL
OpenVX	Graph of image filters (blur, convolve, colour)	Not built (see the OpenVG list)
SDL2 / SDL3	`SDL_Renderer` : blend modes, scaling, render targets, geometry	Partial: scaling and clip. No blend modes, no render targets
Mesa Panfrost	Open driver for Bifrost	Used: bm takes the job descriptors (`v7.xml`) and the shader compiler
Mesa vc4 / v3d	Open driver for VideoCore	Read for the structures and the probes. Linux code is read, never copied

Open points

- Pi Zero W: the 2D scenes `map8`, `sprites`, `stack`, `fade` and `light` were measured once (22 Oct). The `-O3` list for `gfx16.c` is not measured.
- RGB30 with the Mali: only the 3D Bench figures from 25 and 28 Oct exist. The 2D scenes are not measured on the RGB30.
- Flips and alpha each alone, 2D particles and 3D particles on the Pi: not measured.
- **[TO FILL: the Mali-G52 hardware Z-buffer figures, when track G2 ends]**
- **[TO FILL: the RGB30 figures of the four-core 2D bands, which are not built]**

Open points

- **Open points**: the Mali-G52 hardware Z-buffer figures, when track G2 ends
- **Open points**: the RGB30 figures of the four-core 2D bands, which are not built